Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOCHEMISTRY/MICROBIOLOGY)
SEMESTER I EXAMINATION (BACKLOG)
BC702/MI702: CELL BIOLOGY

Date: 04.05.2010
Time: 10.00 AM to 1.00 PM **Total Marks: 70**

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A **Total marks: 20**

Q1. Choose the correct option and put √ mark in front of it:

1. N^+K^+ ATPase **DOES NOT** function directly in
   (a) maintaining membrane potential (b) osmotic regulation
   (c) transport of Na and K ions (d) none of the above
2. The mechanism by which a cell engulfs external fluid is called
   (a) phagocytosis (b) pinocytosis (c) endocytosis (d) none of the above
3. All the following are the functions of golgy body except
   (a) protein sorting (b) vesicular transport
   (c) power house of the cell (d) post translational modification of protein
4. Which of the following correctly matches an organelle with its function
   (a) mitochondria= photosynthesis (b) nucleus= cellular respiration
   (c) lysosome = movement (d) central vacuole = storage
5. Which cell organelle is responsible for production of ATP
   (a) chloroplast (b) nucleus (c) vesicle (d) mitochondrion
6. Cyanide interferes with the cellular aerobic production of ATP by directly influencing the following cell organelle
   (a) lysosome (b) mitochondria (c) ribosomes (d) nucleus
7. The receptors that require physical contact between the cells are
   (a) paracrine (b) endocrine (c) autocrine (d) juxtacrine
8. Signal transduction involves the following
   (a) receptor protein (b) target protein
   (c) intracellular signaling molecules (d) all of the above
9. EGF acts through following type of signaling pathway
   (a) endocrine (b) autocrine (c) paracrine (d) juxtacrine

10. The extracellular matrix can include the following molecular components

(a) glycosaminoglycans (b) glycoproteins (c) laminin (d) all of the above

11. *cdc* mutants are used to study the cell cycle of

(a) *Saccharomyces cerevisieae* (b) *Xenopus laevis*

(c) *Schizosaccharomyces pombe* (d) none of the above

12. The sequence of phases during a cell cycle follows the order

(a) $G1 \rightarrow S \rightarrow G2 \rightarrow M$ (b) $S \rightarrow G1 \rightarrow G2 \rightarrow M$

(c) $G1 \rightarrow M \rightarrow G2 \rightarrow S$ (d) $M \rightarrow S \rightarrow G2 \rightarrow G1$

13. Anaphase promoting complex is a

(a) ubiquitin ligase (b) DNA ligase (c) DNA Polymerase (d) none of the above

14. In the metaphase the sister chromatids are held together by

(a) cohesins (b) condensin (c) adhesins (d) none of the above

15. Mitosis takes place during

(a) G1 phase (b) S phase (c) M phase (d) all of the above

Q2. State True or False:

1. Cholesterol acts as a membrane lipid. **[True] [False]**
2. RBCs in the blood **do not** coagulate under normal physiological conditions. **[True] False]**
3. Corticotropin (ACTH) acts on adrenal cortex. **[True] [False]**
4. Ca^{+2} is **NOT** required for tight junction integrity. **[True] [False]**
5. Voltage gated ion channels have narrow pore and show low ion selectivity. **[True] [False]**

Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY)
SEMESTER II EXAMINATION
BC702/MI702: CELL BIOLOGY

Instructions to the candidates:

1. Attempt all questions
2. Write answers for Section I and Section II of **Part B** in separate answer sheets.

Part B **Total Marks: 50**

Section I

Q.1 (A) Explain the molecular transport mechanism underlying treatment of diarrhea using ORS (mixtures of salt and sugar) (4)

Q.1(B) Write notes on **any TWO** of the following: (6)

(i) Mechanism of H^+ transfer in bacteriorhodopsin
(ii) Role of cholesterol in membrane fluidity and restricted mobility of cell membrane proteins.
(iii) Action potential
(iv) Na^+ K^+ ATPase

Q.2 (A) Discuss the various functions of endoplastic reticulum. (4)

OR

Q.2 (A) Discuss the various functions of golgi bodies. (4)

Q.2 (B) Draw the structures of nucleopore complex and briefly explain its function in eukaryotic cell. (4)

OR

Q.2(B) What is maternal inheritance? Explain the role of mitochondria in maternal inheritance. (4)

Q.2(B) Answer **any one** of the following: (2)

(i) Explain the difference between eukaryotic and prokaryotic cell.
(ii) Justify: mitochondria is the power house of cell.

Section II

Q.3(A) Write note on **any two** of the following: (6)

(i) second messengers (ii) molecular switches

(iii) G-protein linked receptors (iv) extracellular signalling

Q.3(B) Explain **any two** of the following: (4)

(i) Scaffold proteins (ii) Gated ion channels

(iii) Enzyme linked receptors (iv) First messenger

OR

Q.3 (B) Discuss any one of the following: (4)

(i) JAK-STAT transduction mechanism.

(ii) MAPK cascades

Q.4(A) Answer **any one** of the following: (4)

(i) Explain how does controlled degradation of extracellular matrix occur in a mammalian system.

(ii) Explain different types of cell-cell junctions.

Q4. (B) Answer **any two** of the following: (6)

(i) Signals don't require membrane bound receptors. Explain with examples.

(ii) What are proteoglycans? Give the role of hyaluronan in extracellular matrix.

(iii) What are cadherins? Explain the role of cadherins in cellular organization.

Q.5(A) Write a short note on **any two** of the following: (6)

(i) cyclin B (ii) anaphase promoting complex

(iii) condensins (iv) MPF

Q.5(B) Explain briefly **any two** of the following: (4)

(i) cytokinesis (ii) cohesins (iii) G1 CDK complexes